1. Research and Select the Appropriate Generative AI Model

1. Introduction

Generative Artificial Intelligence (Generative AI) refers to AI systems that can create new content from user instructions or prompts. Depending on the model, the generated content can include text, stories, dialogues, descriptions, images, and other creative outputs.

This activity focuses on researching and selecting suitable generative AI models for an application that automatically creates personalized comic stories and illustrations. The selection process begins by understanding the application requirements and matching those requirements with the capabilities of available models.

2. Understand the Project Requirements

The source project, ComicCraft, is a web-based application that accepts user information such as a story prompt, character name, setting, story tone, and art style. It generates a panel-by-panel comic storyline and corresponding illustrations. The generated comic can be previewed through the web interface and exported as a PDF.

Main content requirements

• Structured comic outlines arranged panel-by-panel.

• Full comic narration and character dialogues.

• Detailed scene descriptions and image prompts.

• Comic-style illustrations for each panel.

• Creative, engaging, and visually rich output based on custom prompts.

3. Purpose of Model Selection

Selecting the appropriate model is important because different generative AI models have different strengths. A model suitable for fast structured text may not be the same model used for detailed storytelling or image generation. Therefore, the application can use different models for different stages of the generation process.

4. Research of Candidate Generative AI Models

The source document identifies Google Gemini models for text generation and Hugging Face Diffusers with Stable Diffusion for image generation. The models are evaluated according to the task they need to perform in the ComicCraft workflow.

4.1 Gemini Flash

Gemini Flash is selected for fast and structured generation of comic panel outlines. In the project workflow, it receives the user's story prompt and produces a structured outline that can be used as the foundation for the later storytelling stage.

Main role

• Generate a structured comic outline.

• Organize the story into individual panels.

• Provide a clear structure for subsequent narration generation.

• Support a responsive application workflow through faster generation.

4.2 Gemini Pro

Gemini Pro is selected for detailed and creative comic narration and character dialogue. After the panel outline is prepared, the model expands the outline into a more complete comic-style story.

Main role

• Generate detailed narration for each panel.

• Create engaging character dialogues.

• Expand short outlines into richer story content.

• Follow the requested tone and creative direction.

4.3 Stable Diffusion

Stable Diffusion is selected for generating comic-style illustrations from image prompts. It is used after the text-generation stages so that each panel can receive a corresponding visual representation.

Main role

• Generate images from text-based image prompts.

• Create comic-style visual scenes.

• Support different visual styles requested by users.

• Provide illustrations corresponding to individual story panels.

5. Model Evaluation Criteria

Before final selection, the candidate models should be compared against the requirements of the application. The source document highlights several practical evaluation factors.

5.1 Response Quality

The generated response should be meaningful, coherent, creative, and suitable for the intended application. For a comic generator, the quality of narration and dialogue directly affects the usefulness of the final comic.

5.2 Prompt Adherence

The model should follow the user's instructions accurately. The project allows the user to provide a story prompt, character name, setting, tone, and art style. The generated result should reflect these inputs.

5.3 Richness of Detail

Detailed scene descriptions and storytelling are important for producing an engaging comic. Rich descriptions can also provide better input for the image-generation stage.

5.4 Speed and Generation Performance

Speed is important in an interactive web application. Users expect the system to process their prompts and produce results without unnecessary delay. Image generation performance is also considered because several comic panels may require separate images.

5.5 Model Stability

The model should provide consistent results during repeated testing. Stable behavior helps the application produce predictable outputs for different users and prompts.

5.6 Documentation and Community Support

Availability of documentation, examples, libraries, and community support is useful during implementation and troubleshooting. The source project lists documentation for Gemini, Hugging Face Diffusers, PyTorch, FastAPI, and other technologies used by the application.

6. Model Comparison

Model	Type	Project Task	Key Strength
Gemini Flash	Text	Panel-by-panel outline	Fast and structured generation
Gemini Pro	Text	Narration and dialogue	Detailed and creative text generation
Stable Diffusion	Image	Comic illustrations	Text-to-image comic-style generation

7. Practical Testing Approach

The source document describes practical testing using different prompts to assess storytelling abilities and image-generation performance. A representative prompt can be used to test whether the complete pipeline produces a coherent comic.

Example test prompt

“A brave fox exploring an enchanted forest.”

Expected testing observations

• Whether the system creates a clear multi-panel outline.

• Whether the story remains consistent across panels.

• Whether character dialogue matches the story situation.

• Whether scene descriptions contain useful visual details.

• Whether Stable Diffusion creates illustrations appropriate to the generated prompts.

• Whether changing tone or art style results in an appropriately regenerated comic.

8. Iterative Testing

The project can be tested with different tones and art styles. For example, a user can select a funny tone and comic-book style, after which the pipeline regenerates the outline, story, and images using instructions suited to the selected preferences.

9. Final Generative AI Model Selection

Based on the project requirements and the model evaluation described in the source PDF, the final model combination uses three specialized components. Each model is assigned a specific generation task.

9.1 Selected Model – Gemini Flash

Gemini Flash is selected for generating the structured comic panel outline. Its role is to transform the user's initial story prompt into an organized sequence of panels that can be processed by the next stage.

9.2 Selected Model – Gemini Pro

Gemini Pro is selected for generating detailed narration and engaging character dialogues. It expands the panel outline into the actual textual story content used in the comic.

9.3 Selected Model – Stable Diffusion

Stable Diffusion is selected for generating comic-style illustrations. Image prompts created from the story information are supplied to the image-generation stage to create visuals for the panels.

10. Generation Pipeline

The selected models work as a sequential pipeline. The process starts with the user's input and moves through outline generation, detailed storytelling, image generation, layout creation, and final export.

Step	Process
1	User enters story prompt, character, setting, tone, and art style.
2	Gemini Flash generates a structured comic outline.
3	Gemini Pro expands the outline into narration and dialogue.
4	Image prompts are prepared for individual panels.
5	Stable Diffusion generates comic-style illustrations.
6	Text and images are combined into the comic layout.
7	The completed comic can be previewed and exported as PDF.

11. Expected Outcome and Conclusion

The selected combination of generative AI models allows the application to perform different stages of comic creation efficiently. Gemini Flash handles the structured planning stage, Gemini Pro handles detailed creative writing, and Stable Diffusion handles visual illustration generation.

This model-selection approach supports the main objective of ComicCraft: transforming a simple user prompt into a complete, visually engaging comic. The generated output can contain panel-by-panel narration, character dialogue, scene illustrations, and a final downloadable PDF.

12. Benefits of the Selected Approach

• Separates text generation and image generation into appropriate AI tasks.

• Supports personalized comic generation based on user inputs.

• Provides structured storytelling before detailed narration is generated.

• Creates corresponding illustrations for individual comic panels.

• Supports different tones and visual styles.

• Makes comic creation accessible to users without advanced drawing skills.

• Provides an end-to-end workflow from prompt entry to PDF export.

13. Final Summary

The research and selection activity demonstrates that model choice should be based on the actual requirements of the application. For the ComicCraft project, the selected combination is Gemini Flash for structured comic outlines, Gemini Pro for detailed narration and character dialogues, and Stable Diffusion for comic-style illustrations. Together, these models provide the text and visual generation capabilities required by the project.

14. Source Note

This six-page document is adapted from the uploaded “ComicCraft - AI Comic Story Creator using Gemini Models” PDF, especially the sections covering Project Workflow, Milestone 1, Activity 1.1, model evaluation, and final model selection.